

Tracks & Problem Statements for Hack Helix

T 1

Climate Tech & Sustainability

Solutions must handle messy, incomplete real-world environmental data - not clean CSV demos

Problem 01

Embodied Carbon Negotiator for Construction Procurement

Construction material procurement happens before buildings are built, but the carbon cost is locked in at that moment. Design a system that compares material options (steel grade, cement type, insulation) by embodied carbon across the full supply chain - not just the product spec sheet - and generates a procurement shortlist that hits a target carbon budget without exceeding cost margins.

Problem 02

Groundwater Depletion Attribution Engine

Multiple industries draw from the same aquifer without visibility into cumulative impact. Build a system that attributes groundwater stress to industrial sectors using satellite surface subsidence data, published usage permits, and rainfall anomaly records - and models which sector's reduction would have the highest aquifer recovery impact.

Problem 03

Industrial Heat Waste Recovery Optimizer

Factories and data centers release enormous amounts of waste heat that nearby facilities (greenhouses, cold-storage, district heating) could absorb - but no matching system exists. Build a proximity-aware platform that ingests industrial heat output profiles and local demand schedules to identify viable heat-sharing pairs, accounting for pipe distance, temperature compatibility, and seasonal mismatch.

T 2

Cybersecurity & Adversarial Robustness

Build systems that hold up when the attacker knows your system exists

Problem 01

Adversarial Prompt Firewall for LLM-Integrated Products

As companies embed LLMs into their products, attackers craft inputs that hijack the model's behavior - leaking system prompts, bypassing filters, or executing indirect injections through documents and URLs. Build a modular, pluggable firewall layer that sits between user input and any LLM backend, classifying adversarial prompt patterns in real time without blocking legitimate edge-case queries.

Problem 02

Insider Threat Behavioral Baseline System

Most security tools look for known attack signatures. Build a system that establishes a behavioral baseline per user - login times, file access sequences, data transfer volumes - and flags statistically significant deviations that could indicate a compromised or malicious insider, without relying on rule-based filters.

Problem 03

Open-Source Dependency Exploit Propagation Mapper

When a CVE is published for an open-source library, thousands of downstream projects are affected but most maintainers don't know. Build a system that traces transitive dependency trees across public repositories (npm, PyPI, Maven) to identify affected projects, estimate exposure severity by usage context, and draft automated patch-request issue text for maintainers.

T 3

Intelligent Infrastructure & Real-World Systems

Physical-world problems where software alone isn't enough - sensors, pipelines, and decisions under uncertainty

Problem 01

Multi-Modal Last-Mile Delivery Failure Predictor

Last-mile delivery failures cost logistics companies 15-20% of route cost. Build a system that ingests historical delivery records, address geocodes, time-of-day patterns, and weather overlays to predict failure probability per stop before dispatch - and resequences the route and contact strategy accordingly.

Problem 02

Cold Chain Integrity Auditor for Pharmaceutical Distribution

Temperature excursions during vaccine and drug transport are common but underreported. Design a system that reconstructs the likely temperature history of a shipment from sparse sensor logs, GPS traces, and ambient climate data for the route - and computes a product integrity score that flags batches requiring inspection, even when sensor coverage is incomplete.

Problem 03

Electricity Theft Detection in Distribution Networks

Power distribution companies lose 20-30% of electricity to theft and meter tampering in India, but audits are manual and infrequent. Build an anomaly detection system that identifies suspicious consumption patterns at the transformer or feeder level by comparing expected load from connected meters against actual grid draw.

Problem 04

Structural Health Index from Passive Vibration Sensing

Bridges and older buildings degrade silently until failure. Build a system that processes vibration time-series data from cheap accelerometers mounted on a structure to compute a structural health index - detecting changes in resonant frequency that correlate with material fatigue, without requiring calibrated instruments or engineers on-site.

T 4

Human Augmentation & Assistive Systems

Extend what people can perceive, decide, and do - especially those underserved by existing tools

Problem 01

Real-Time Sign Language Synthesis from Natural Speech

Deaf individuals in India are underserved because ISL (Indian Sign Language) differs structurally from English - it has its own grammar, spatial structure, and non-manual markers. Build a system that converts continuous spoken Hindi or English into ISL gesture sequences using a 3D avatar, handling sentence reordering and topic-comment structure, not just word-for-word sign lookup.

Problem 02

Cognitive Fatigue-Aware Interface Adapter

People with ADHD, post-concussion syndrome, or chronic fatigue experience drastically reduced interface usability during high-fatigue states. Build a system that infers real-time cognitive load from passive input signals (typing rhythm, scroll behavior, cursor jitter) and progressively simplifies the interface - without the user having to configure anything.

Problem 03

Legal Document Simplifier with Risk Surface Mapping

Contracts are written to protect the party with the lawyer. Build a system that takes an uploaded agreement (rental, employment, loan, terms of service) and produces a plain-language summary alongside a risk surface map - identifying clauses that are unusual, one-sided, or waive significant rights - ranked by potential user impact.

T 5

Trustworthy AI & Responsible Systems

The hard problems in AI aren't the models - they're what happens when models meet reality

Problem 01

Hallucination Audit Trail for LLM-Generated Documents

When LLMs produce reports or research briefs, they sometimes fabricate citations, statistics, or events. Build a system that automatically traces every factual claim in an LLM-generated document back to a verifiable source - and produces an audit report distinguishing verified claims, unverified but plausible claims, and hallucinations, with confidence levels attached.

Problem 02

Dataset Bias Cartography Tool

ML models trained on biased datasets produce biased outputs - but most bias audits only check protected attributes after training. Build a pre-training dataset analysis tool that maps demographic, linguistic, geographic, and contextual attribute distributions across a dataset - surfaces underrepresented slices - and generates a bias risk report before a model is trained on it.

Problem 03

Model Decision Contestation Interface

Automated systems deny loans, flag job applications, and moderate content - with no mechanism for users to meaningfully contest the decision. Design a structured contestation interface that allows affected individuals to submit counter-evidence, forces the model to re-evaluate under the new information, and produces an explainable delta - showing exactly what changed and whether the decision flipped or held.